| | |
|---|---|
| **Provably nonempty** | <spaces>, <comment>, <comma>, <ident-startchar>, <ident-anychar>, <ident>, <size>, <hexvalue>, <decvalue>, <value>, <concrete-value>, <abstract-value>, <data-value>, <data-label>, <data-section>, <instr-ldr>, <instr-any>, <instr-code>, <register-number>, <register-alias>, <register>, <constant>, <deref-offset>, <deref-reg>, <local-label>, <eqlabel>, <operand>, <comma-operand>, <operands>, <instruction>, <inline-data>, <inline-label>, <print-width>, <register-slice>, <print-register>, <print-list>, <print-directive>, <directive>, <text-contents>, <text-section>, <arm> |
| **Possibly empty** | <xspaces>, <lb>, <linebreak>, <data-contents> |

| spaces | | examples |
|---|---|---|
| ␣ | | |
| ␣␣␣␣␣␣␣␣ | | |

| comment | | examples |
|---|---|---|
| //␣comment | | |
| //␣comment¤ | | |
| // | | |
| //¤ | | |

| comma | | examples |
|---|---|---|
| , | | |
| ␣␣␣␣,␣ | | |

| lb | | examples |
|---|---|---|
| ␣␣␣␣␣␣␣␣␣␣¤<br>␣␣␣␣␣␣¤<br>␣␣␣␣␣␣␣ | | |

| linebreak | | examples |
|---|---|---|
| ␣␣␣␣␣¤<br>¤ | | |
| //␣comment¤<br>¤<br>//␣eh | | |

| linebreak | | counterexamples |
|---|---|---|
| ¤<br>␣␣␣␣␣␣␣x | | **Parsing error:** The parser did not consume the entire input.<br>1 \| ¤<br>~\| Surplus characters on next line<br>\| Valid <linebreak><br>Hint: halted due to the following:<br>2 \| ␣␣␣␣␣␣␣x<br>^ Can't match string "<br>"<br>While trying to parse: <linebreak> → <lb>. |
| //␣comment¤<br>␣␣␣x | | **Parsing error:** The parser did not consume the entire input.<br>1 \| //␣comment¤<br>~~~~~~~~~~~\| Surplus characters on next line<br>\| Valid <linebreak><br>Hint: halted due to the following:<br>2 \| ␣␣␣x<br>^ Can't match string "<br>"<br>While trying to parse: <linebreak> → <lb>. |
| //␣comment¤<br>¤<br>␣␣␣x | | **Parsing error:** The parser did not consume the entire input.<br>2 \| ¤<br>~\| Surplus characters on next line<br>\| Valid <linebreak><br>Hint: halted due to the following:<br>3 \| ␣␣␣x<br>^ Can't match string "<br>"<br>While trying to parse: <linebreak> → <lb>. |
| //␣comment¤<br>␣␣␣x | | **Parsing error:** The parser did not consume the entire input.<br>1 \| //␣comment¤<br>~~~~~~~~~~~\| Surplus characters on next line<br>\| Valid <linebreak><br>Hint: halted due to the following:<br>2 \| ␣␣␣x<br>^ Can't match string "<br>"<br>While trying to parse: <linebreak> → <lb>. |

| ident-startchar | | examples |
|---|---|---|
| f | | f |
| A | | A |

| ident-startchar | | counterexamples |
|---|---|---|
| `_` | | **Parsing error:** The input does not match the expected format.<br>`1 \| _`<br>`^` Regex '`[A-Z]`' does not match<br>While trying to parse: `<ident-startchar>`. |

| ident-anychar | | examples |
|---|---|---|
| `A` | | A |
| `0` | | 0 |
| `x` | | x |
| `_` | | _ |

| ident | | examples |
|---|---|---|
| `LD_foo` | | `(lab: "LD_foo")` |
| `main` | | `(lab: "main")` |
| `loop0` | | `(lab: "loop0")` |

| ident | | counterexamples |
|---|---|---|
| `0var` | | **Parsing error:** The input does not match the expected format.<br>`1 \| 0var`<br>`^` Regex '`[A-Z]`' does not match<br>While trying to parse: `<ident>` → `<ident-startchar>`. |

| size | | examples |
|---|---|---|
| `.byte` | | `(size: 1)` |
| `.word` | | `(size: 4)` |
| `.hword` | | `(size: 2)` |

| size | | counterexamples |
|---|---|---|
| `.bte` | | **Parsing error:** The input does not match the expected format.<br>`1 \| .bte`<br>`^` Can't match string "`word`"<br>While trying to parse: `<size>`. |

| hexvalue | | examples |
|---|---|---|
| `0x45` | | `(hex: "45")` |
| `0xdead` | | `(hex: "dead")` |

| hexvalue | | examples |
|---|---|---|
| `0xffffffff` | | `(hex: "ffffffff")` |

| hexvalue | | counterexamples |
|---|---|---|
| `0xz` | | **Parsing error:** The input does not match the expected format.<br>`1 \| 0xz`<br>`^ Regex ‘[0-9a-fA-F]’ does not match`<br>While trying to parse: `<hexvalue>`. |

| decvalue | | examples |
|---|---|---|
| `42` | | `(int: "42")` |
| `1000` | | `(int: "1000")` |
| `0` | | `(int: "0")` |

| decvalue | | counterexamples |
|---|---|---|
| `45x` | | **Parsing error:** The parser did not consume the entire input.<br>`1 \| 45x`<br>`~~\|^ Surplus characters`<br>`\| Valid <decvalue>`<br>Hint: halted due to the following:<br>`1 \| 45x`<br>`^ Regex ‘[0-9]’ does not match`<br>While trying to parse: `<decvalue>`. |

| value | | examples |
|---|---|---|
| `0x44` | | `(hex: "44")` |
| `420` | | `(int: "420")` |
| `mask` | | `(lab: "mask")` |

| value | | counterexamples |
|---|---|---|
| `0var` | | **Parsing error:** The parser did not consume the entire input.<br>`1 \| 0var`<br>`~\|^^^ Surplus characters`<br>`\| Valid <value>`<br>Hint: halted due to the following:<br>`1 \| 0var`<br>`^ Regex ‘[0-9]’ does not match`<br>While trying to parse: `<value>` → `<decvalue>`. |

| concrete-value | | examples |
|---|---|---|
| `.byte␣0x42` | | `(size: 1, hex: "42")` |
| `.word␣mask` | | `(size: 4, lab: "mask")` |

| concrete-value | | counterexamples |
|---|---|---|
| `.bte␣0x42` | | **Parsing error:** The input does not match the expected format.<br>`1 \| .bte␣0x42`<br>`^ Can't match string "word"`<br>While trying to parse: `<concrete-value>` → `<size>`. |
| `.byte␣45x` | | **Parsing error:** The parser did not consume the entire input.<br>`1 \| .byte␣45x`<br>`~~~~~~~~\|^ Surplus characters`<br>`\| Valid <concrete-value>`<br>Hint: halted due to the following:<br>`1 \| .byte␣45x`<br>`^ Regex '[0-9]' does not match`<br>While trying to parse: `<concrete-value>` → `<value>` → `<decvalue>`. |

| abstract-value | | examples |
|---|---|---|
| `.skip␣5` | | `(size: 5)` |

| abstract-value | | counterexamples |
|---|---|---|
| `skip␣3` | | **Parsing error:** The input does not match the expected format.<br>`1 \| skip␣3`<br>`^ Can't match string ".skip"`<br>While trying to parse: `<abstract-value>`. |
| `.skip␣0x42` | | **Parsing error:** The parser did not consume the entire input.<br>`1 \| .skip␣0x42`<br>`~~~~~~~\|^^^ Surplus characters`<br>`\| Valid <abstract-value>`<br>Hint: halted due to the following:<br>`1 \| .skip␣0x42`<br>`^ Regex '[0-9]' does not match`<br>While trying to parse: `<abstract-value>` → `<decvalue>`. |

| data-value | | examples |
|---|---|---|
| `.byte␣0x42` | | `(size: 1, hex: "42")` |
| `.skip␣8` | | `(size: 8)` |

| data-label | | examples |
|---|---|---|
| `a:` | | `(lab: "a")` |
| `B:` | | `(lab: "B")` |

| data-contents | | examples |
|---|---|---|
| `A:¤`<br>`␣␣B:␣¤`<br>`␣␣.word␣0x64¤`<br>`␣␣␣␣␣␣␣¤`<br>`␣␣.byte␣0x42¤`<br>`¤`<br>`␣␣¤`<br>`foo:␣.hword␣42¤`<br>`bar:␣baz:␣.byte␣1␣.byte␣2␣.byte␣3` | | `(`<br>`  (lab: "A"),`<br>`  (lab: "B"),`<br>`  (size: 4, hex: "64"),`<br>`  (size: 1, hex: "42"),`<br>`  (lab: "foo"),`<br>`  (size: 2, int: "42"),`<br>`  (lab: "bar"),`<br>`  (lab: "baz"),`<br>`  (size: 1, int: "1"),`<br>`  (size: 1, int: "2"),`<br>`  (size: 1, int: "3"),`<br>`)` |

| data-contents | | counterexamples |
|---|---|---|
| `.byte␣1␣c` | | **Parsing error:** The parser did not consume the entire input.<br>`1 \| .byte␣1␣c`<br>`~~~~~~~~\|^` Surplus characters<br>`\|` Valid `<data-contents>`<br>Hint: halted due to the following:<br>`1 \| .byte␣1␣c`<br>`^` Can't match string ":"<br>While trying to parse: `<data-contents>` → `<data-label>`. |

| data-section | | examples |
|---|---|---|
| `␣␣.data¤`<br>`foo:␣.word␣0x42¤`<br>`bar:␣.word␣0x43` | | `(`<br>`  data: (`<br>`  (lab: "foo"),`<br>`  (size: 4, hex: "42"),`<br>`  (lab: "bar"),`<br>`  (size: 4, hex: "43"),`<br>`  ),`<br>`)` |

| instr-code | | examples |
|---|---|---|
| `ldr` | | `(instr: "ldr", size: 4)` |
| `add` | | `(instr: "add")` |
| `lsl` | | `(instr: "lsl")` |
| `ldrh` | | `(instr: "ldr", size: 2)` |

| instr-code | | examples |
|---|---|---|
| `b` | | `(instr: "b")` |

| register-number | | examples |
|---|---|---|
| `r0` | | `(reg: 0)` |
| `r9` | | `(reg: 9)` |
| `r12` | | `(reg: 12)` |

| register-alias | | examples |
|---|---|---|
| `sp` | | `(reg: 13)` |
| `lr` | | `(reg: 14)` |
| `pc` | | `(reg: 15)` |

| register | | examples |
|---|---|---|
| `r8` | | `(reg: 8)` |
| `sp` | | `(reg: 13)` |
| `lr` | | `(reg: 14)` |
| `r0` | | `(reg: 0)` |

| constant | | examples |
|---|---|---|
| `#4` | | `(int: "4")` |
| `#0x1` | | `(hex: "1")` |

| deref-offset | | examples |
|---|---|---|
| `,␣r1` | | `(reg: 1)` |
| `,␣#3` | | `(int: "3")` |

| deref-reg | | examples |
|---|---|---|
| `[r1]` | | `(deref: ((reg: 1),))` |
| `[␣r2␣␣]` | | `(deref: ((reg: 2),))` |
| `[r1,␣r2]` | | `(deref: ((reg: 1), (reg: 2)))` |
| `[r1␣,␣#1]` | | `(deref: ((reg: 1), (int: "1")))` |
| `[r1,␣r2,␣r3]` | | `(deref: ((reg: 1), (reg: 2), (reg: 3)))` |

| deref-reg | | examples |
|---|---|---|
| `[r1,␣r2,␣#2]` | | `(deref: ((reg: 1), (reg: 2), (int: "2")))` |

| local-label | | examples |
|---|---|---|
| `.LD_foo` | | `(lab: ".LD_foo")` |

| operand | | examples |
|---|---|---|
| `lr` | | `(reg: 14)` |
| `#1` | | `(int: "1")` |
| `[␣r1␣,␣#2␣]` | | `(deref: ((reg: 1), (int: "2")))` |
| `.LD_xx` | | `(lab: ".LD_xx")` |
| `=mask` | | `(eq: "mask")` |
| `loop0` | | `(lab: "loop0")` |

| operands | | examples |
|---|---|---|
| `lr,␣#1` | | `((reg: 14), (int: "1"))` |
| `r0␣,␣r0␣,␣[r1,␣#2]` | | `(`<br>`  (reg: 0),`<br>`  (reg: 0),`<br>`  (deref: ((reg: 1), (int: "2"))),`<br>`)` |

| instruction | | examples |
|---|---|---|
| `ldr␣r0,␣[r1]` | | `(`<br>`  instr: "ldr",`<br>`  size: 4,`<br>`  ops: ((reg: 0), (deref: ((reg: 1),))),`<br>`)` |
| `add␣r0,␣r1,␣r2` | | `(instr: "add", ops: ((reg: 0), (reg: 1), (reg: 2)))` |
| `sub␣r1,␣#1` | | `(instr: "sub", ops: ((reg: 1), (int: "1")))` |
| `lsl␣r1,␣#8` | | `(instr: "lsl", ops: ((reg: 1), (int: "8")))` |

| inline-data | | examples |
|---|---|---|
| `.LD_xx:␣.word␣x` | | `(lab: "x", size: 4)` |
| `.LD_xx:␣¤`<br>`␣␣.byte␣0x42` | | `(lab: ".LD_xx", size: 1, hex: "42")` |

| inline-label | | examples |
|---|---|---|
| `main:` | | `(tag: "main")` |

| print-width | | examples |
|---|---|---|
| `/8` | | 8 |
| `/16` | | 16 |

| register-slice | | examples |
|---|---|---|
| `[:]` | | `(start: auto, len: auto)` |
| `[1:]` | | `(start: (int: "1"), len: auto)` |
| `[16:8]` | | `(start: (int: "16"), len: (int: "8"))` |

| print-register | | examples |
|---|---|---|
| `r0` | | `(reg: 0)` |
| `r0[:8]` | | `(reg: 0, start: auto, len: (int: "8"))` |
| `r0[16:32]` | | `(reg: 0, start: (int: "16"), len: (int: "32"))` |

| print-list | | examples |
|---|---|---|
| `r0,␣r1,␣r2` | | `((reg: 0), (reg: 1), (reg: 2))` |

| print-directive | | examples |
|---|---|---|
| `print/8␣:␣r0` | | `(print: (width: 8, regs: ((reg: 0),)))` |
| `print␣:␣r1,␣r2` | | `(print: (width: auto, regs: ((reg: 1), (reg: 2))))` |

| directive | | examples |
|---|---|---|
| `@␣print/8:␣r0` | | `(print: (width: 8, regs: ((reg: 0),)))` |

| text-contents | | examples |
|---|---|---|
| `main:¤`<br>`␣␣ldr␣r0,␣[r1]¤`<br>`␣␣add␣r0,␣#1␣//`<br>`␣test¤`<br>`loop0:¤`<br>`␣␣eor␣r0,␣r1,␣r2¤`<br>`.LD_data:␣.word␣data` | | `(`<br>`  (tag: "main"),`<br>`  (`<br>`  instr: "ldr",`<br>`  size: 4,`<br>`  ops: ((reg: 0), (deref: ((reg: 1),))),`<br>`  ),`<br>`  (instr: "add", ops: ((reg: 0), (int: "1"))),`<br>`  (tag: "loop0"),`<br>`  (instr: "eor", ops: ((reg: 0), (reg: 1), (reg: 2))),`<br>`  (lab: "data", size: 4),`<br>`)` |

| text-section | | examples |
|---|---|---|
| ␣␣.text¤<br>¤<br>main:¤<br>␣␣mov␣r0,␣#1␣//<br>␣test¤<br>␣␣add␣r0,␣r0,␣r0 | | ( text: ( (tag: "main"), (instr: "mov", ops: ((reg: 0), (int: "1"))), (instr: "add", ops: ((reg: 0), (reg: 0), (reg: 0))), ), ) |

| arm | | examples |
|---|---|---|
| ␣␣.data¤<br>A:␣.byte␣0x64¤<br>␣␣␣.word␣0x95¤<br>␣␣␣.hword␣45¤<br>¤<br>␣␣¤<br>␣␣.text¤<br>main:␣//<br>␣main␣function¤<br>␣␣ldr␣r0,␣.LD_A¤<br>␣␣ldr␣r0,␣[r6]¤<br>␣␣ldrb␣r0,␣[r6]¤<br>␣␣ldrh␣r0,␣[r5,␣#1]␣//<br>␣yay¤<br>␣␣mov␣r5,␣r6¤<br>␣␣mvn␣r5,␣r6¤<br>␣␣mov␣r5,␣#0x1¤<br>␣␣add␣r5,␣r6,␣#1¤<br>␣␣␣¤<br>␣␣lsl␣r5,␣#1¤<br>␣␣lsr␣r5,␣#1␣//<br>␣do␣some␣stuff¤<br>␣␣//<br>␣testing␣comments¤<br>␣␣eor␣r3,␣r4,␣r5¤<br>␣␣orr␣r3,␣r4,␣r5¤<br>␣␣and␣r3,␣r4,␣r5␣¤<br>␣␣¤<br>␣␣¤<br>.LD_A:␣.word␣A¤<br>␣␣ | | ( data: ( (lab: "A"), (size: 1, hex: "64"), (size: 4, hex: "95"), (size: 2, int: "45"), ), text: ( (tag: "main"), ( instr: "ldr", size: 4, ops: ((reg: 0), (lab: ".LD_A")), ), ( instr: "ldr", size: 4, ops: ((reg: 0), (deref: ((reg: 6),))), ), ( instr: "ldr", size: 1, ops: ((reg: 0), (deref: ((reg: 6),))), ), ( instr: "ldr", size: 2, ops: ((reg: 0), (deref: ((reg: 5), (int: "1")))), ), (instr: "mov", ops: ((reg: 5), (reg: 6))), (instr: "mvn", ops: ((reg: 5), (reg: 6))), (instr: "mov", ops: ((reg: 5), (hex: "1"))), ( instr: "add", ops: ((reg: 5), (reg: 6), (int: "1")), ), (instr: "lsl", ops: ((reg: 5), (int: "1"))), (instr: "lsr", ops: ((reg: 5), (int: "1"))), (instr: "eor", ops: ((reg: 3), (reg: 4), (reg: 5))), (instr: "orr", ops: ((reg: 3), (reg: 4), (reg: 5))), (instr: "and", ops: ((reg: 3), (reg: 4), (reg: 5))), (lab: "A", size: 4), ), ) |

| parsing | | |
|---|---|---|
| 112 | 112 | 0 |